

Vision-Language Models, and Whether You Can Trust Them

Robert Pless · Trustworthy AI · CSCI 6510 / EMSE 6992

One geometry under everything

- Images and text are embedded into one shared space (CLIP space).
- Classify, search, and generate are all “measure distance in that space.”
- **Every capability lives in that geometry — and so does every way trust breaks.**
- We will teach the geometry once, then return to it five times, each time darker.

Deep Learning is Not Just for Text (or images) or Video by themselves

- Scrape all of the images from the internet and their associated text/captions
- Learn the relationship between text and images
- Train a model to create new images that are aligned with particular captions

Stable Diffusion 2.1 Demo

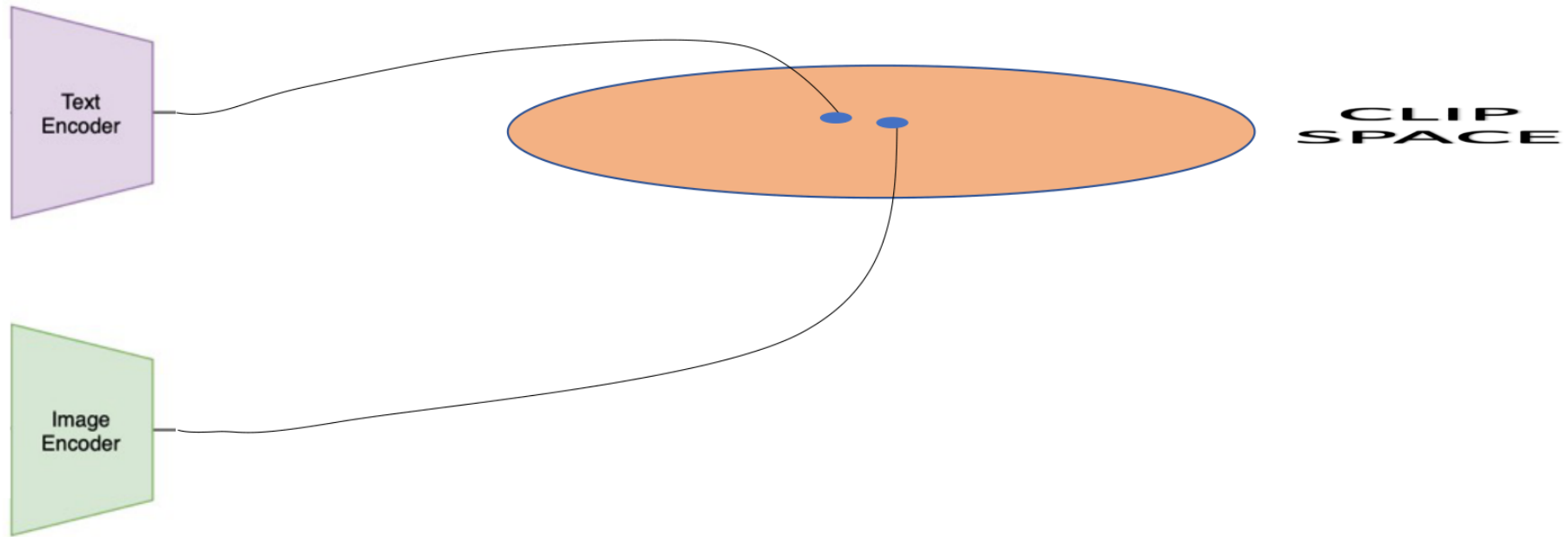
Stable Diffusion 2.1 is the latest text-to-image model from StabilityAI. [Access Stable Diffusion 1 Space here](#)

For faster generation and API access you can try [DreamStudio Beta](#).

A computer fighting in a war

Generate image





CLIP (Contrastive Language Image Pre-Training), is the technical basis that allows current approaches to link images and text.

Trained on 5 billion pairs of (image, caption), CLIP has learned to embed images and related text to nearby locations (in a high-dimensional vector space).

This is surprisingly powerful.

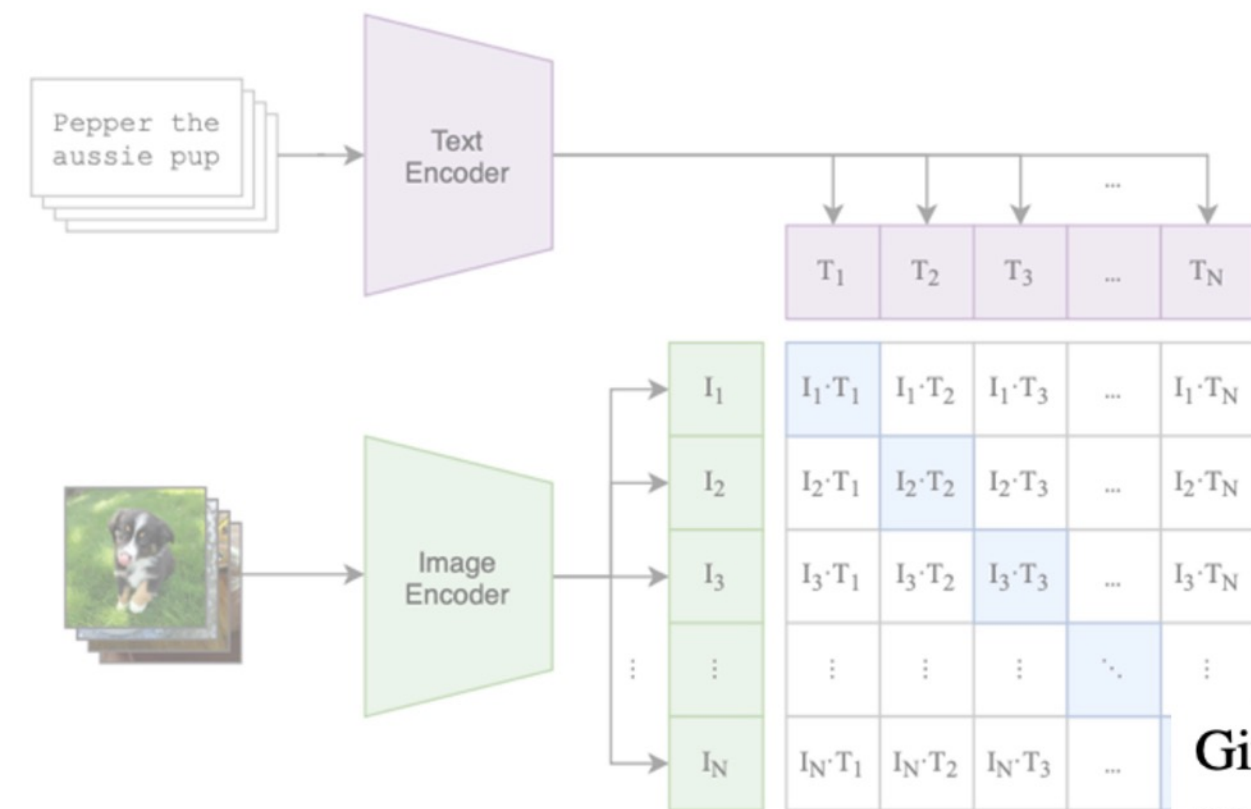
Correction: Original CLIP trained on ~400M pairs (WIT-400M). “5B” is LAION-5B, the open reproduction — not the original model.

Learning Transferable Visual Models From Natural Language Supervision

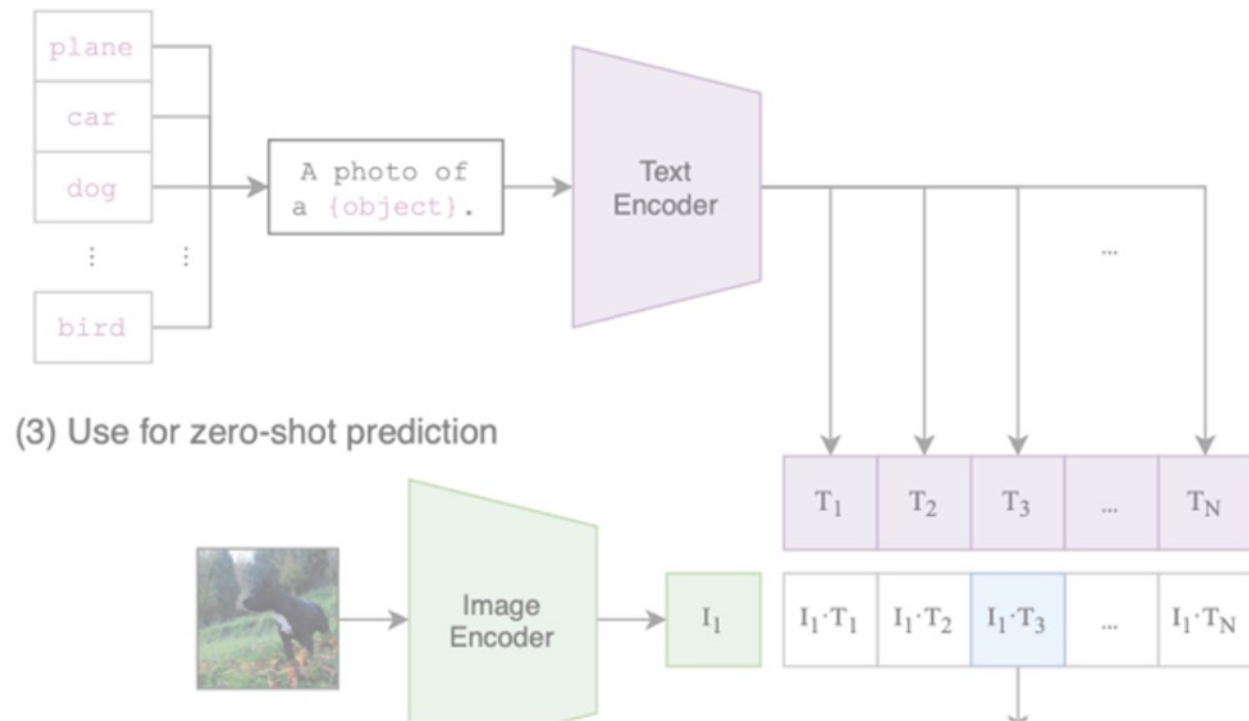
Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹

CLIP:
Contrastive
Language
Image
Pre-training

(1) Contrastive pre-training



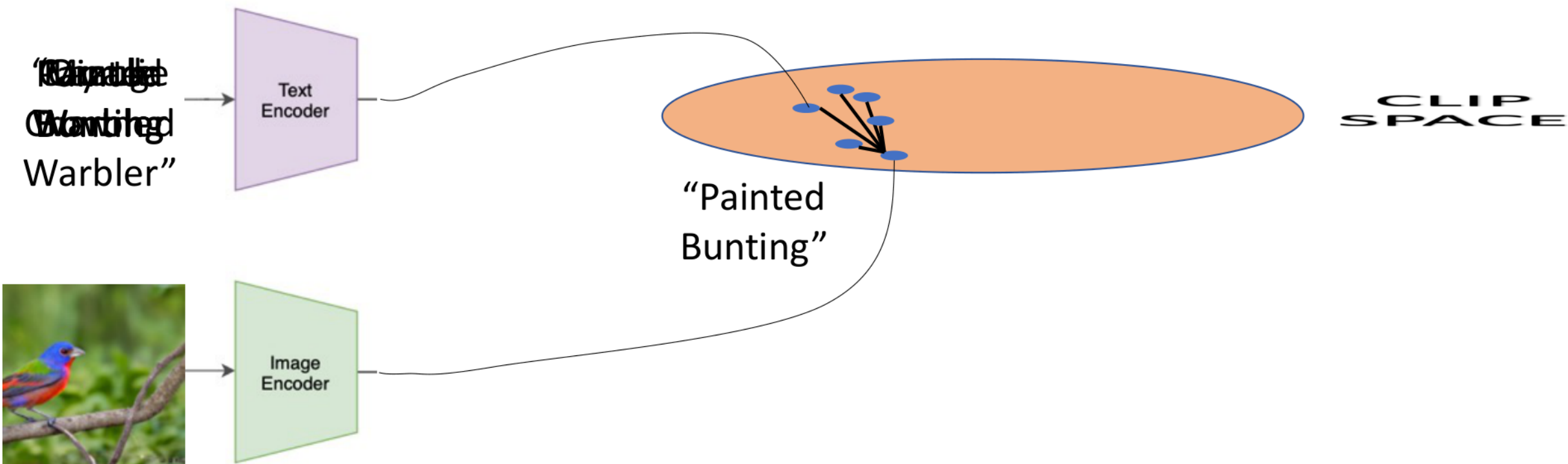
(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

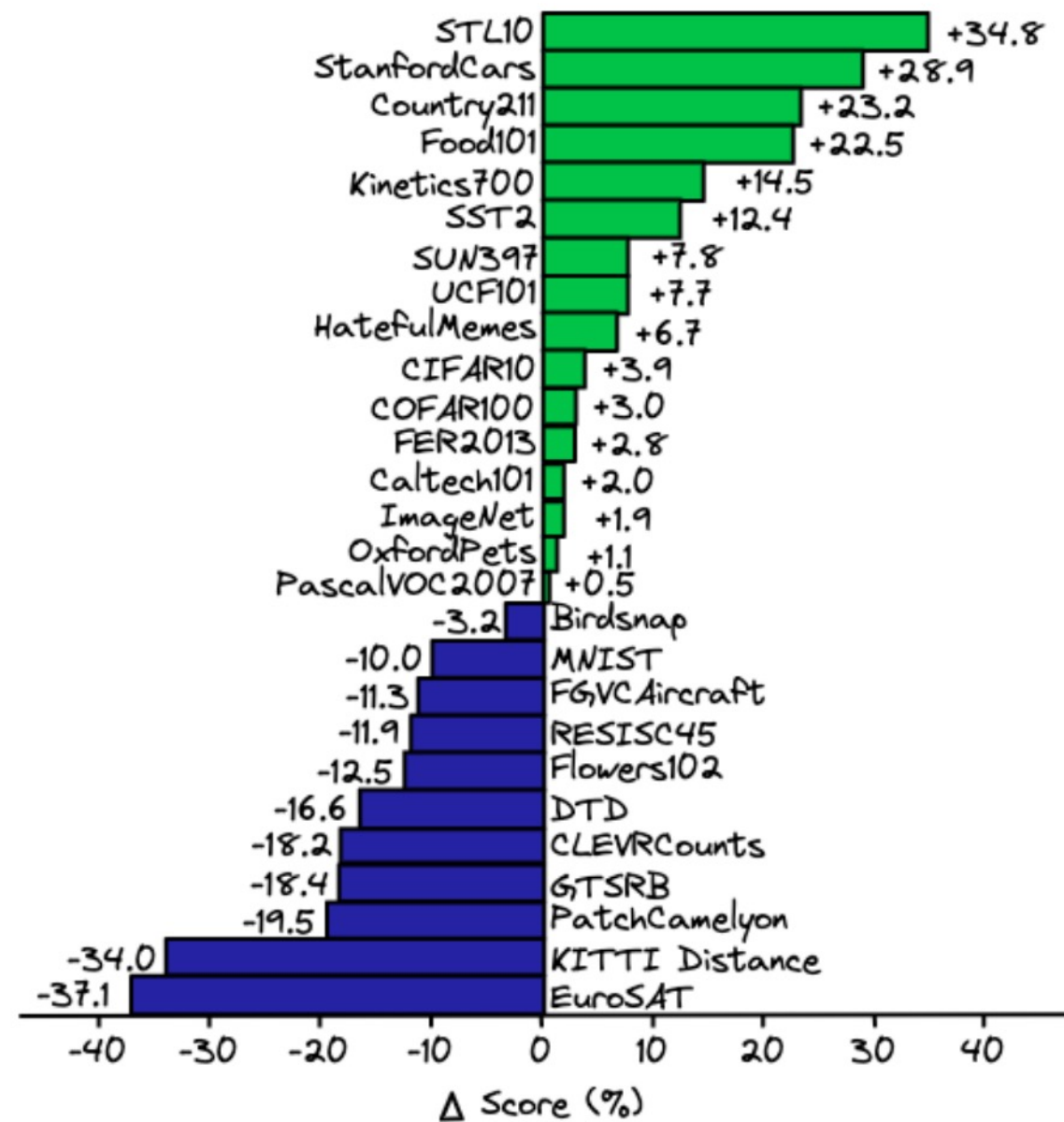
Figure 1. Summary of our approach. While standard image models require some label, CLIP jointly trains an image encoder and a text encoder on image-text pairs. At test time the learned text encoder synthesizes a target dataset’s classes.

Given a batch of N (image, text) pairs, CLIP is trained to predict which of the $N \times N$ possible (image, text) pairings across a batch actually occurred. To do this, CLIP learns a multi-modal embedding space by jointly training an image encoder and text encoder to maximize the cosine similarity of the image and text embeddings of the N real pairs in the batch while minimizing the cosine similarity of the embeddings of the $N^2 - N$ incorrect pairings. We optimize a symmetric cross entropy loss over these similarity scores.



This is surprisingly powerful. It also makes very quick, easy image classifiers

CLIP based classifiers outperform hand-tuned, specially trained classifiers for many, many datasets



If you have a problem for which you need image classification, you no longer need to talk to Computer Vision faculty. Now any good undergrad can make you a classifier for exactly the classes you care about.

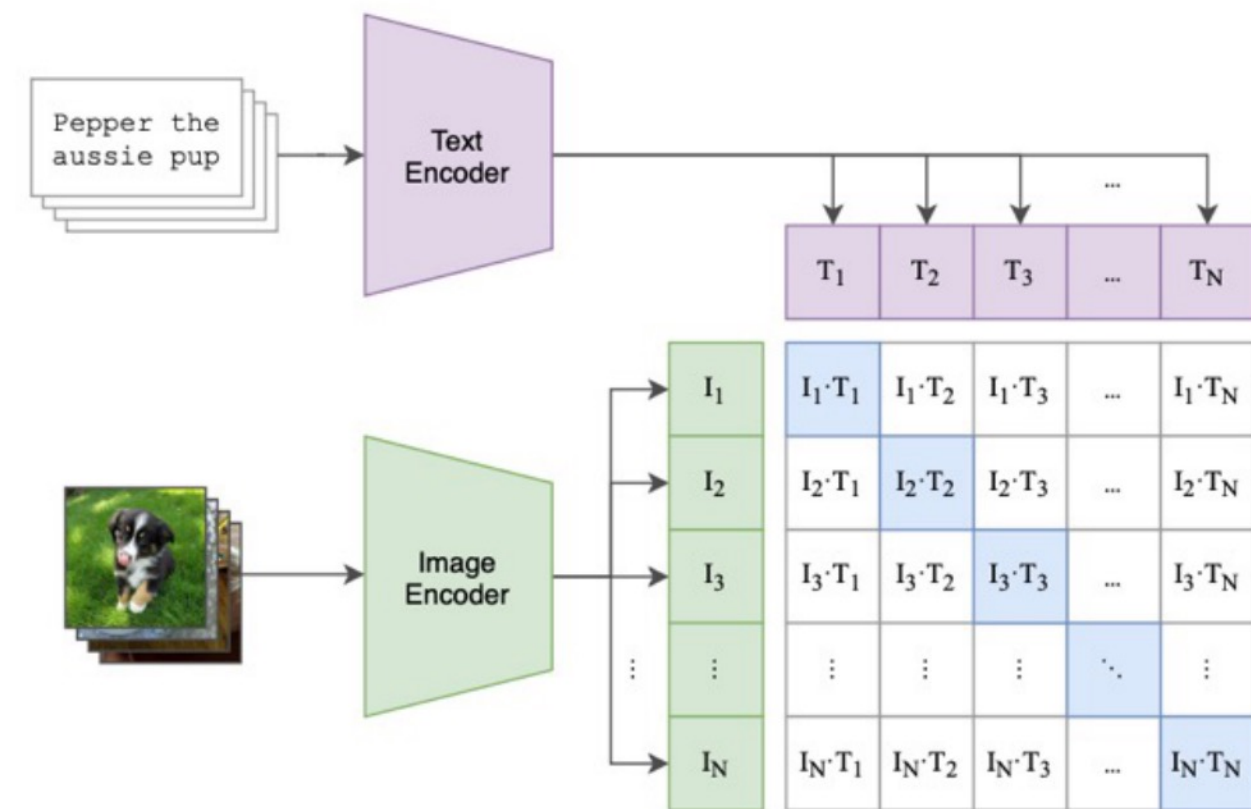
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ML Background Questions:

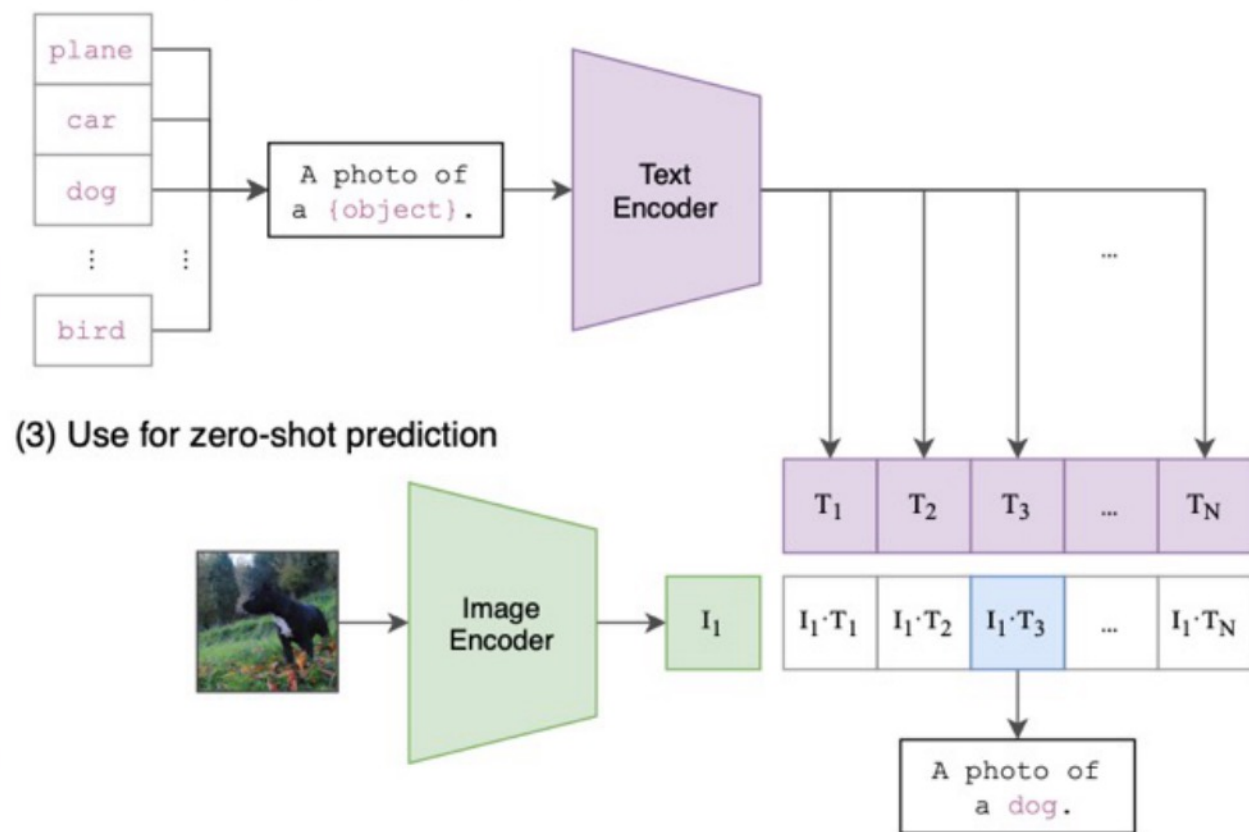
“What is the SOTA algorithm?”

- “What is one-shot learning?”
- “What is few-shot learning?”
- “What is zero-shot learning?”

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset’s classes.

Module 2 — Can you trust it?

- The turning point: “Will CLIP Zero-Shot?” — your work with Xiaotong Liu, Kevin Robbins, Yu Wu.
- The mechanics half ended on “...but are they good?” This module is the answer.
- The trust question: how do you know a zero-shot classifier works before you have any labeled data?

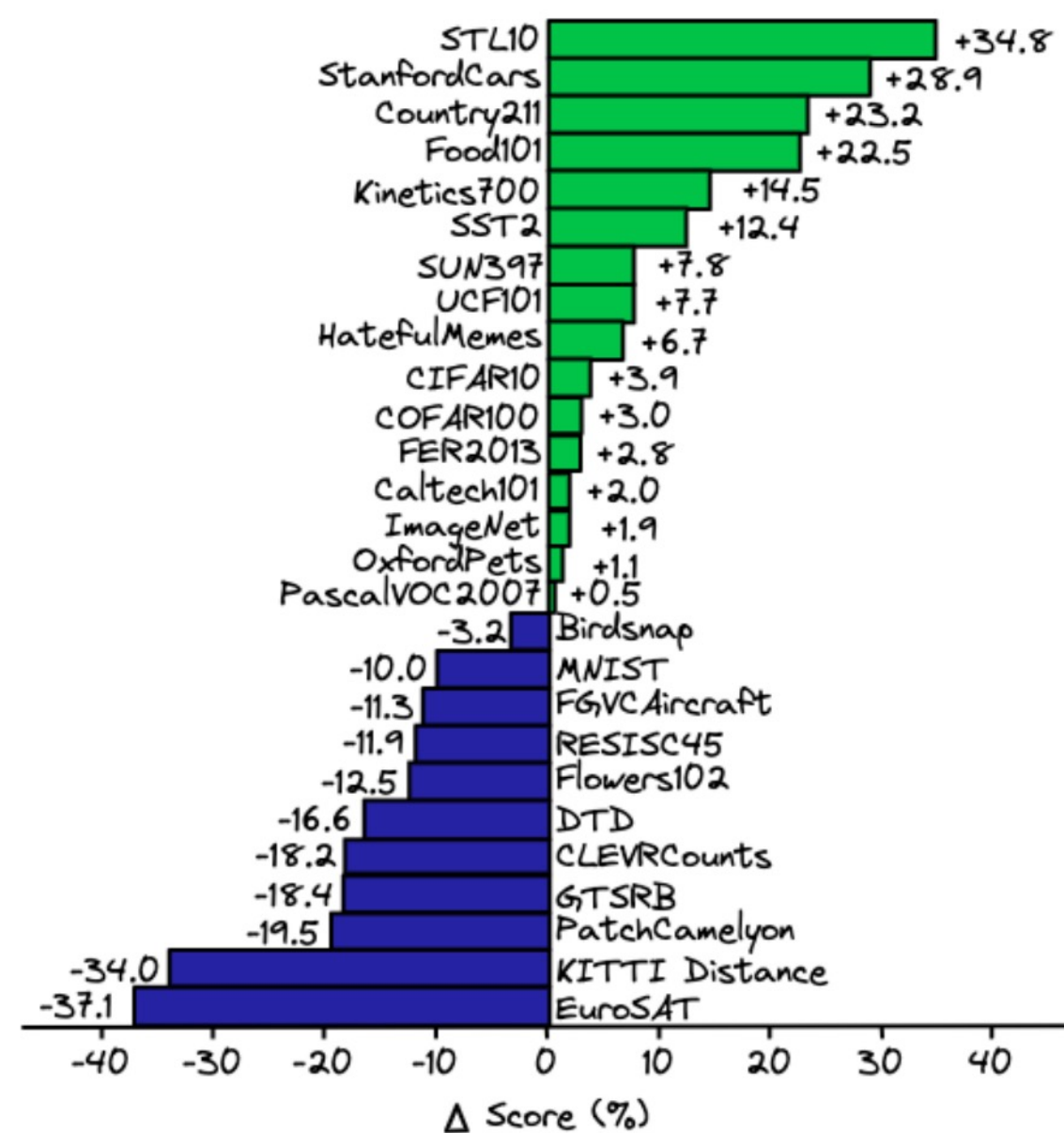
... But you might still want to know how well it works for *your* problem domain...

It might fail because:

1. you are making up nonsense categories
2. CLIP doesn't have a good embedding of your categories
3. CLIP embedding is ambiguous (photo of "black knight" could relate to a person, a chess piece or Batman)
4. ??? --- sometimes Deep Learning has confusing error modes.

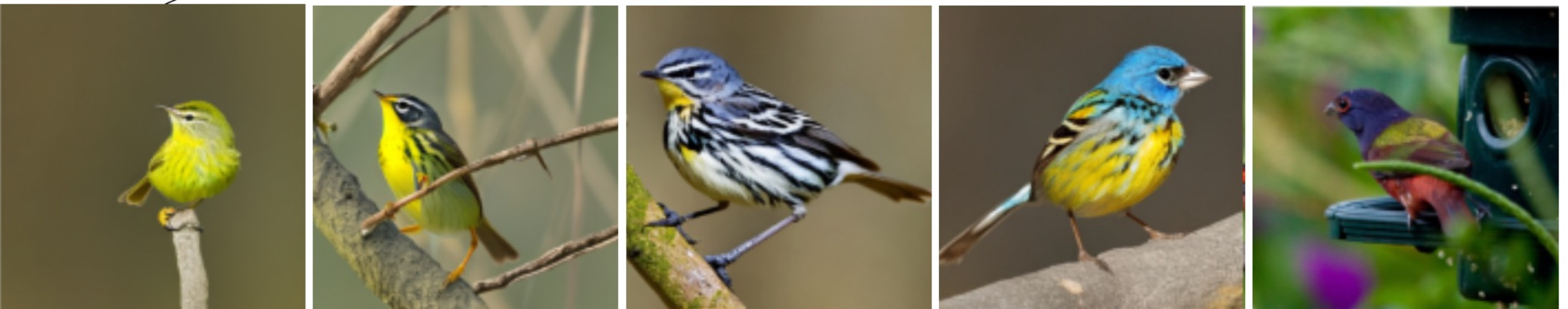
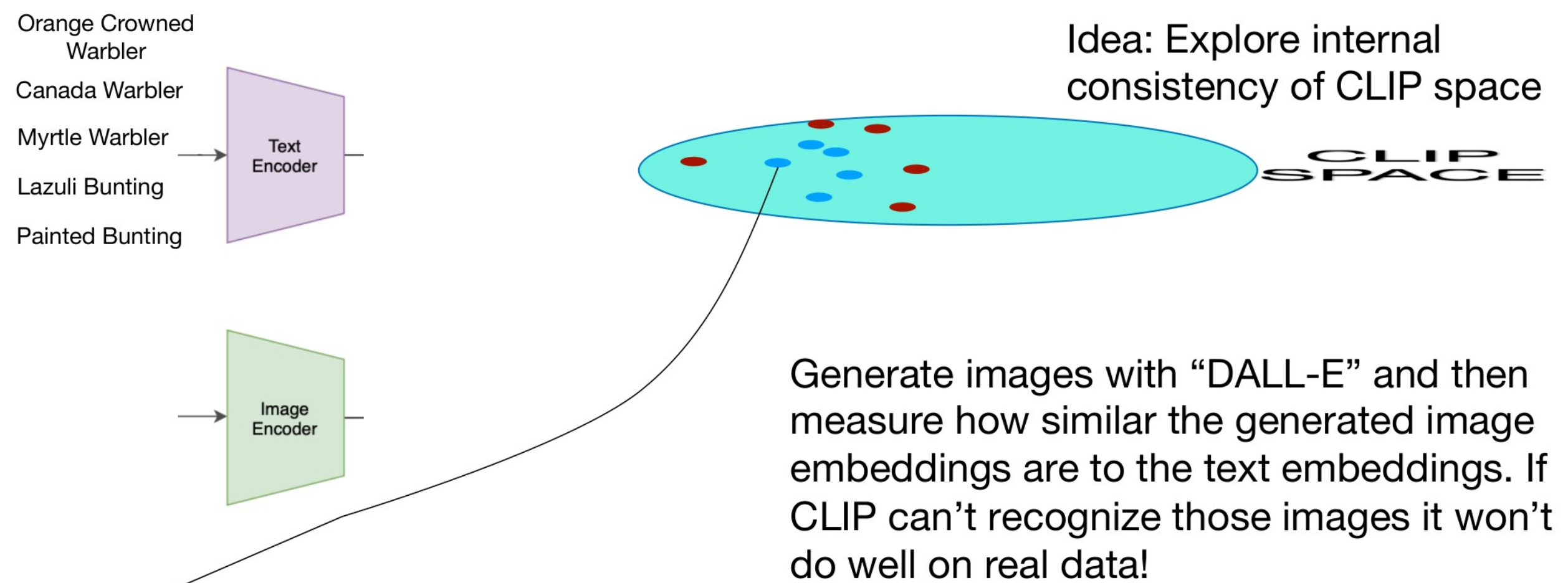
If you have labelled data in your domain, you can just test how well CLIP works.

But if you don't have labelled data, then the undergraduate student has to be very good to find the right data to test on, and that might take a while

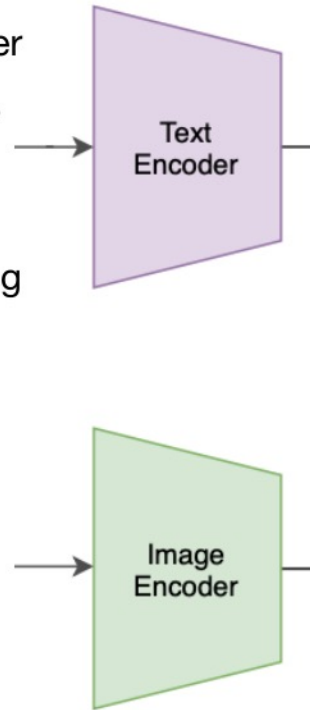


Question: Without any data from the problem domain, how well can you quickly and automatically predict the accuracy of CLIP based image classification?

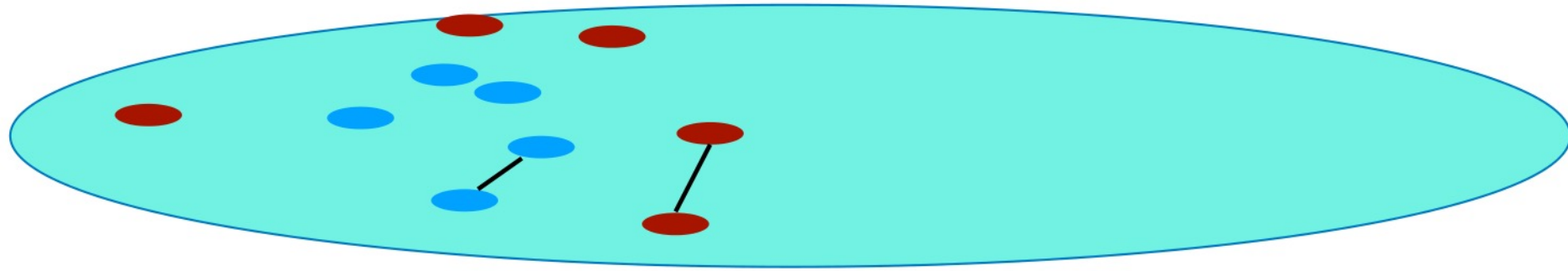
But if you don't have labelled data, then the undergraduate student has to be very good to find the right data to test on, and that might take a while



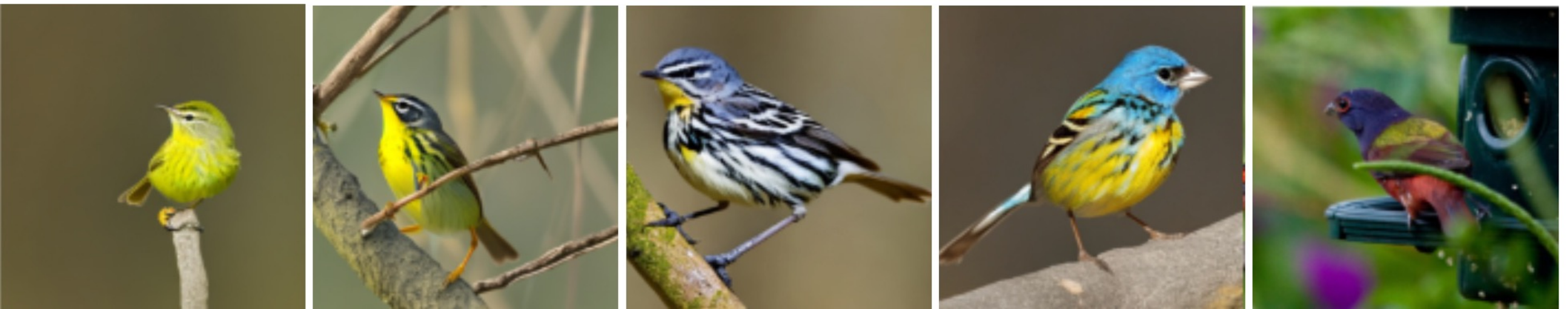
Orange Crowned
Warbler
Canada Warbler
Myrtle Warbler
Lazuli Bunting
Painted Bunting



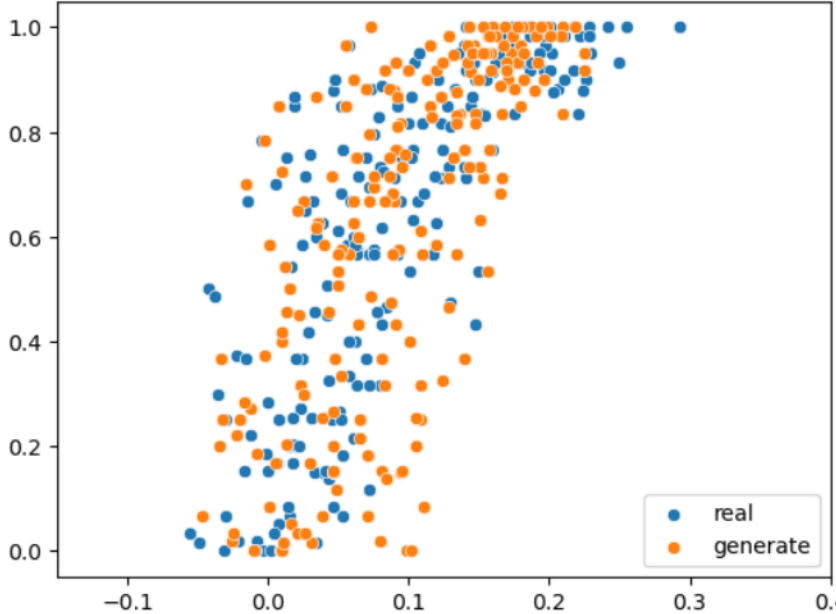
Idea: Refine analysis of CLIP space



Measure if “text difference vector between similar classes” is parallel to the “image difference vector between those classes”



CUB-200, classification accuracy for each of then 200 classes, as predicted by real images, and by generated images



	Real Image	Generated Image
CUB-200	0.98 0.82	0.67 0.72

Closest text to each image
“Refined Score”



domestic sheep

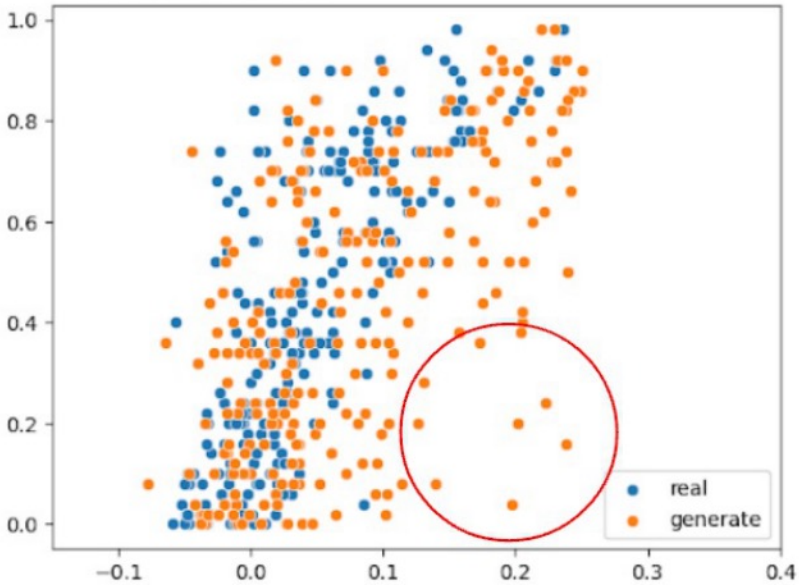
High consistency score, but actually low Recall@1

Generator is creating accurate and consistent images, but real world imagery is much more varied.



Wild Boar

Mammal:



Summary

We can partially predict the accuracy with which CLIP will recognize visual categories.

Some of the measured error in real datasets comes from challenges in defining the classes – future work will explore improving prediction accuracy by better defining those classes.

Module 3 — Bias is geometry

- Same space, same operations — now the neighbors are people.
- **Every capability from Modules 1–2 has a harm twin in the same geometry.**

Bias in CLIP: Why It Matters

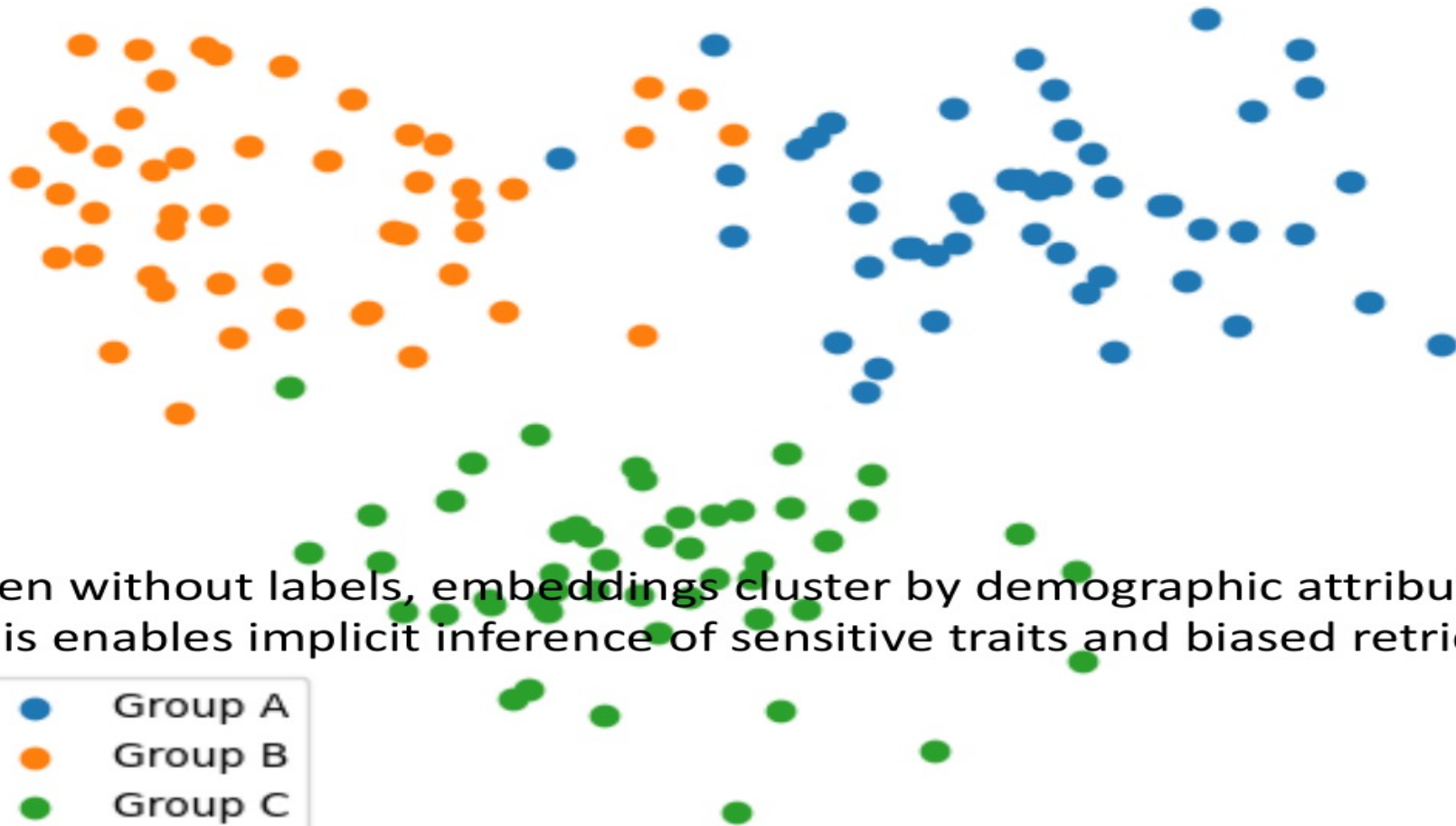
- CLIP learns similarity structure from image–text co-occurrence on the web.
- Bias is encoded in embedding geometry, not explicit labels.
- Used for retrieval, zero-shot classification, filtering → bias propagates.
- Small geometric shifts can change rankings and decisions.

Training Data: LAION and Web-Scale Bias

- CLIP-style models rely on massive web-scraped datasets (e.g., LAION).
- Captions reflect stereotypes, Western norms, and skewed visibility.
- Some identities appear frequently; others only in narrow contexts.
- Filtering removes extreme cases, not subtle correlations.

t-SNE: Demographic Clustering in CLIP Embeddings

Illustrative t-SNE of CLIP Image Embeddings
(Demographic Clustering)



Even without labels, embeddings cluster by demographic attributes.
This enables implicit inference of sensitive traits and biased retrieval.

Observed Failure Modes

- Occupations cluster by gender and race.
- Attributes (e.g., "professional", "criminal") align with demographics.
- Sensitive traits inferred via nearest neighbors.
- Bias affects ranking, recall, and zero-shot accuracy.

What Can Be Done (and What Can't)

- Measure bias in embedding space, not just outputs.
- Audit retrieval results, not only classification accuracy.
- Use calibration, re-ranking, or task-specific adapters.
- No full fix: web-scale pretraining bakes in societal bias.

2021: the apple that read “iPod”

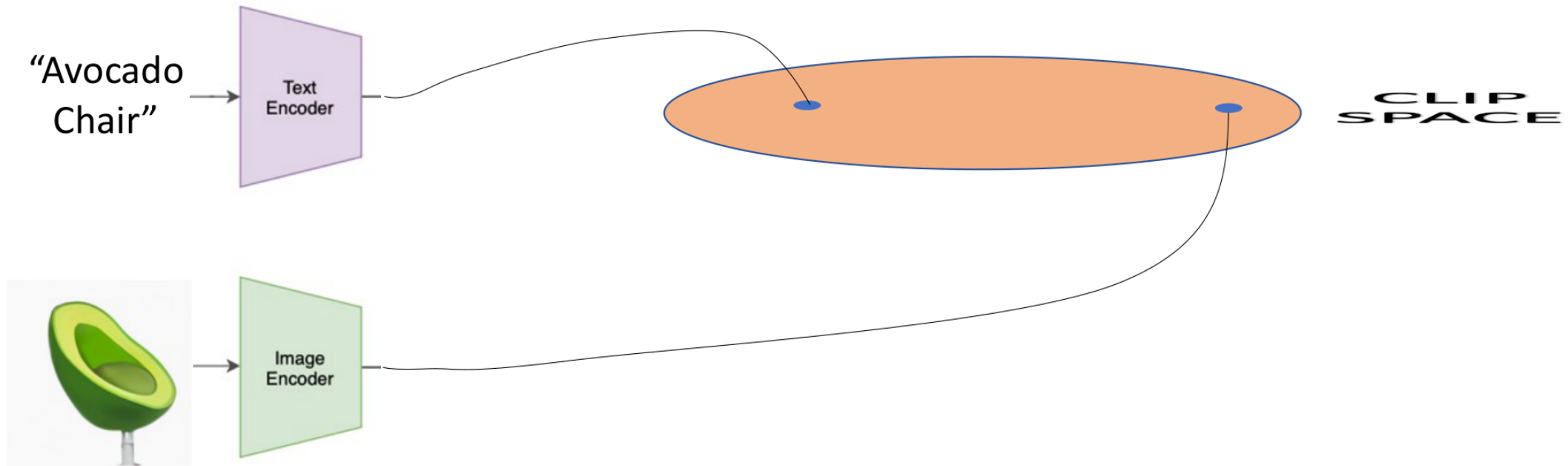
- Tape a handwritten “iPod” label on an apple, and original CLIP calls it an iPod.
- The model reads text in the image and lets it override the pixels.
- **The early lesson: “recognition” is not understanding.**
- *Screenshot to add: Goh et al., “Multimodal Neurons” (Distill, 2021) — the labeled-apple figure.
<https://distill.pub/2021/multimodal-neurons/>*

2026: the same trick now hijacks agents

- It didn't get fixed — it graduated into visual / typographic prompt injection.
- Instructions hidden in an image that a VLM-driven agent then obeys.
- **~64% attack success under stealth constraints vs. GPT-4V, Claude 3, Gemini, LLaVA (2026).**
- Treated in the literature as a systemic property of multimodal alignment, not a patchable bug.
- *Screenshot to add: CSA image-prompt-injection note (2026).*
<https://labs.cloudsecurityalliance.org/research/csa-research-note-image-prompt-injection-multimodal-llm-2026/>

This is the lethal trifecta, arriving through a camera

- Same trust-boundary problem as your agentic lectures: untrusted input the model treats as instructions.
- Now it rides in on a physical sign, package, or screen in a camera-equipped agent's field of view.
- **Demonstrated against driving assistants (January 2026).**
- *Screenshot to add: Physical Prompt Injection Attacks on LVLMs, arXiv 2601.17383 — a physical-sign example.*



Application 2: Dall-E. (“Generating images from Text”)

- Your text is mapped into CLIP space to create an image target.
- Then a random starting image is created and mapped into CLIP space.
- Then gradient descent is used to update pixel values to change the image --- so that the CLIP embedding moves closer to the target

Correction: Pixel-space gradient descent = CLIP-guided generation (VQGAN+CLIP lineage). DALL·E 2 is unCLIP: a diffusion prior + diffusion decoder.

Shockingly Good

Prompt: Luke Skywalker ordering a burger and fries from the Death Star canteen.



Shockingly Good

Prompt: Black adventure van, geared out, apocalyptic, dystopian van, realistic, roof rack, in dark abandoned neighborhood, van geared out



Shockingly Good (*sometimes*)

Why would this happen?

The CLIP model mapping text to images is trained with images and captions --- of all possible objects, what is most likely to be in an image with a caption that says “no elephant”?



Safety, Bias & Provenance

- *How do CLIP's biases propagate through unCLIP, and what safeguards are realistic?*
- Bias in latents → biased images
- Data curation, prompt filters
- Watermarks/provenance (e.g., C2PA)
- Detection models (imperfect)

And now it's the law

- The EU AI Act's transparency rules took effect 2 August 2026.
- AI-generated or manipulated content, including deepfakes, must be disclosed.
- Generative outputs must be marked in a machine-readable way.
- Connects image generation to a dated, checkable governance fact — and back to your Lecture One EU AI Act slide.

In-class exercise

In small groups, take one prompt (from the CLIP / unCLIP reading questions):

- Noisy web data & bias: how do we keep the model from just absorbing the web's biases?
- Negation & compositionality: “a photo of a car that is not red” — why is this hard?
- Prompt sensitivity: how much does wording change zero-shot accuracy, and can we systematize it?
- Safety, bias & provenance: how do CLIP's biases propagate through generation, and what safeguards are realistic?

One geometry, all of it

- The same embedding gave us zero-shot classifiers, semantic search, and image generation —
- and gave us biased retrieval, injectable agents, and undetectable fakes.
- **Trusting a VLM means reasoning about that geometry, not the demo.**